MUHAMMAD QAIM ALI

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer with 1.5+ years of production experience delivering 100+ engagements across generative AI, agentic AI, and autonomous automation infrastructure for enterprise and direct clients in e-commerce, fintech, healthcare, and manufacturing. Architects end-to-end AI and data platforms — from RAG-based retrieval and vector-database optimization to fine-tuned vision-language models, autonomous agents, real-time streaming pipelines, and resilient automation infrastructure — with hands-on MLOps and AWS deployment experience.

TECHNICAL SKILLS

Programming: Python, C++, SQL **ML/DL:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LoRA/PEFT, Model Optimization

NLP & LLMs: LangChain, spaCy, NLTK, RAG, Vector Databases (Qdrant, pgvector), Prompt Engineering, AI Agents

Computer Vision: YOLOv8, ResNet, DICOM Processing, 3D Vision-Language Models, OpenCV

Automation & Browser Engineering: Selenium, Puppeteer, undetected-chromedriver, GoLogin, proxy rotation, CAPTCHA-bypass (2Captcha), anti-detection systems

MLOps & Data Engineering: MLflow, DVC, Apache Airflow, Docker, Docker Compose, CI/CD (GitHub Actions)

Big Data & Distributed Processing: Apache Spark (Structured Streaming, PySpark), Apache Kafka, Hive, Hive Metastore, Impala, Delta Lake, Avro, Parquet

Cloud (AWS): EC2, RDS, S3, IAM, VPC, ASG, ELB, CloudWatch, EMR **Databases:** PostgreSQL (pgvector), Neo4j, Qdrant, MongoDB

APIs & Backend: FastAPI, RESTful APIs, JWT, RBAC, Async SQLAlchemy **Tools:** VS Code, Git, Docker, Jupyter

EXPERIENCE

Independent Contractor Remote

AI & Automation Engineering — Direct Clients (Selected Engagements) *Mar 2024 – Present*

- Delivered 100+ engagements for direct clients spanning generative AI, agentic AI, autonomous automation infrastructure, and applied data engineering across e-commerce, fintech, healthcare, and social media, collaborating closely with client stakeholders from requirements through delivery and sustaining a 5-star satisfaction record.
- **ReplyPilot** | *Python, GoLogin, Generative AI* — Deployed an agentic AI reply system fusing generative and templated logic to autonomously decide how each inbound message should be handled, resolving ~80% of messages across 5 social platforms (X, TikTok, Facebook, Instagram, Snapchat) and cutting response time from hours to under a minute.
- **LeadForge Pipeline** | *Selenium, pandas, Google Maps API, Zoho, Taptify* — Architected a fully autonomous, end-to-end lead-generation and reputation-management automation system: Selenium-driven sourcing, pandas-based cleaning and deduplication, Google Maps logo enrichment with automated upscaling, dynamic landing-page generation, and white-label review-funnel provisioning with auto-generated QR codes — slashing per-client onboarding time by ~70% with zero manual handoffs.
- **BoltScraper** | *Python, Selenium, Custom GUI* — Engineered a scalable, production-grade lead-generation automation platform with a custom GUI, automating the scraping and structuring of industry-specific leads across 10+ verticals — scaling addressable lead volume ~3x and lifting accuracy ~30%.
- **GhostProvision Suite** | *Puppeteer, Selenium, undetected-chromedriver, 2Captcha* — Built resilient, autonomous automation infrastructure with proxy rotation and CAPTCHA bypass, provisioning 500+ accounts at 95%+ success, plus a Cloudflare-evading real-time data scraper sustaining ~98% uptime.
- **AutoTrader Engine** | *NinjaTrader, Python, Real-Time Market Data* — Built a fully autonomous algorithmic trading system with real-time market monitoring, boosting execution speed 10x over manual trading at ~99% uptime.

Systems Limited Islamabad, Pakistan

AI/ML Engineering Intern *Jul 1, 2026 – Aug 28, 2026*

- Cut vector DB query latency by ~99.9% (2,000ms → ~3ms) by diagnosing a transport-layer misconfiguration causing a search-latency floor in a self-hosted benchmark, then switching the client from REST to gRPC.

- Benchmarked Qdrant, FAISS, and Weaviate across efficiency and BEIR accuracy metrics, selecting the production-grade vector DB configuration for agent query retrieval.
- Deployed Help and Research AI agents to validate the recommended vector DB configuration against production-representative retrieval workflows.

Arbisoft Islamabad, Pakistan

Machine Learning Intern *Jul 2025 – Sep 2025*

- Collaborated with the engineering team to build retrieval components for *Alithia Knowledge*, a RAG-based knowledge-retrieval platform serving enterprise clients in production.
- Cut query latency by **20%** by migrating the knowledge base from Neo4j to Qdrant vector DB, validating embedding pipelines for semantic consistency with zero data loss.

FAST-NUCES Faisalabad, Pakistan

Teaching Assistant *Aug 2023 – Jun 2026*

- Collaborated with course instructors to deliver lab and tutorial instruction for Operating Systems, Expository Writing, and English Composition & Comprehension.

KEY ACHIEVEMENTS

PAB MRI: Multimodal AI for Clinical Report Generation | *PyTorch, LoRA, DICOM, MLflow* | Link 2025 – 2026

- Set out to ease the radiology reporting bottleneck by converting raw MRI scans directly into structured clinical narratives. Engineered a 3D Vision-Language Model pipeline with full IRB-compliant DICOM preprocessing on a proprietary dataset of 4,000+ patient scans, applying LoRA fine-tuning within a full MLOps workflow (DVC, MLflow) for real-time inference — targeting expert-level diagnostic accuracy on a task that traditionally requires a trained radiologist.

Ecommerce Data Pipeline | *Kafka, Spark, Delta Lake, Hive, Impala, AWS* | Link 2026

- A client needed both real-time and historical visibility into their e-commerce data without running two separate systems. Built a unified platform — self-hosted Kafka ingesting Avro-serialized events through Spark Structured Streaming for real-time analytics, plus scheduled Spark batch jobs for historical reporting — landing in partitioned Delta Lake tables on S3 behind a shared Hive Metastore, queryable via Impala and Spark SQL and optimized with broadcast joins and Bloom filter indexing.

Intelligent Clinic System | *FastAPI, PostgreSQL, Scikit-learn, RAG, Docker, AWS* | Link 2026

- Built to give multi-tenant healthcare providers secure patient-data isolation alongside intelligent risk detection and staff support. Architected a production-grade API with JWT, RBAC, and Row-Level Security to keep tenant data isolated, trained an Isolation Forest model on 5,000 vital records that achieved 100% precision/recall flagging anomalies, and integrated a RAG pipeline (LLaMA 3, nomic-embed-text) so staff could query records conversationally — containerized with Docker Compose and shipped with 49 automated tests via GitHub Actions CI/CD.

Manufacturing Defect Detection | *PyTorch, ResNet-50, Computer Vision* | Link 2025

- Manufacturing QA needed reliable visual defect detection, but real defect examples were rare relative to normal parts, so a naive classifier would miss them. Designed a ResNet-50 defect-detection system with weighted loss strategies specifically to counter the class imbalance, building a full augmentation-to-deployment pipeline around it — improving F1-score by 18% over the baseline model.

ADDITIONAL AUTOMATION PROJECTS

- **Instagram Engagement Bot** — Anti-detection automation with randomized delays and coordinate-aware interaction, reducing account-flagging incidents by ~90% versus naive automation approaches.
- **Vinted Marketplace Scraper** — Resilient scraper bypassing anti-bot defenses (proxy rotation, user-agent spoofing, CAPTCHA handling), sustaining a ~95% scrape success rate across paginated listings.
- **University Application Bot** — Multi-step form-automation system with retry and error-handling logic, processing hundreds of applications per batch at a ~95% submission success rate.
- **YouTube Channel Provisioning Bot** — Bulk channel creation and metadata configuration from structured account lists, cutting per-channel setup time by ~85%.

EDUCATION

FAST National University of Computer & Emerging Sciences (NUCES) Faisalabad, Pakistan

B.S. Artificial Intelligence — CGPA: 3.14 *2022 - 2026*

- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, NLP, Computer Vision, MLOps, Cloud Computing, Data Structures & Algorithms, Database Systems
- Silver Medalist — 1st Semester (Fall 2022); DEAN's List — 7th Semester (Fall 2025); DEAN's List — 8th Semester (Spring 2026)